MLE Syllabus

PLSC 606J, Fall 2026

Instructor

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- office hours: M 1:30-3:30pm and by appt.

Course details

- Fall 2026
- Wednesday
- 11:30am-2:30pm
- Bing SSEL - LNG 332

Seminar Description

This 4 credit hour seminar is a survey of maximum likelihood methods and their applications to empirical political questions. It presumes students have a detailed and intuitive knowledge of least squares, probability theory, basic skills in scalar and matrix algebra, and a basic understanding of calculus. The course will deal mainly in understanding the principles of maximum likelihood estimation, under what conditions we move away from least squares, and what particular models are appropriate given observed data. The seminar will focus on application and interpretation of ML models and linking theory to statistical models. The course emphasizes coding and data viz in R.

The class meets one time per week for three hours. My office hours are designed to be homework help hours where I'll work in the grad lab with any of you who are working on the exercises. The most productive pathway for this class is for you to get in the habit of working together, and those office hours are a good time for this.

Course Purpose

This seminar fulfills the advanced quantitative methods requirement in the Ph.D. curriculum. The method of maximum likelihood underlies a majority of quantitative models in Political Science; this class teaches students to be astute consumers of such models, and how to implement and interpret ML models. These are crucial skills for dissertations in Political Science, and for producing publishable quantitative research.

Learning Objectives

Students will encounter an array of maximum likelihood models in this course. By the end of the course, students will have mastered the theory of maximum likelihood sufficient to write and program likelihood functions in R; they will be able to choose, estimate, and interpret appropriate models, model specifications, and model evaluation tools given their data; and they will be able to produce sophisticated quantities of interest (e.g. predicted probabilities, expected values, confidence intervals) via a variety of techniques including simulation and end point transformation. Students will also be able to present model findings verbally and graphically.

R Resources

There are lots of good, free R resources online. Here are a few:

- Modern Statistics with R
- R for Data Science
- R Markdown: The Definitive Guide
- R Graphics Cookbook
- Advanced R
- R Markdown Cookbook
- Data Science:A First Introduction
- The Big Book of R

Class Meetings, Office Hours, Assignments

The course will meet this fall entirely in-person in the Social Science Experiment Lab on Wednesdays 11:30am-2:30pm.

Office hours are Mondays 1:30pm-3:30pm. I'll hold these in the grad work room to help with your assignments. For an appointment, email me and we'll sort out a time.

This syllabus, slides, etc. can be found on the course website. Assignments will be posted weekly to Brightspace. All assignments should be turned in on Brightspace - please submit ::

- PDFs generated from *LaTeX* or R Markdown (Quarto).
- annotated R scripts.
- where necessary, data.

Assignments should be instantly replicable - running the code file should produce all models, tables, plots, etc.

Reading

The reading material for the course is important because it often demonstrates application of various MLE models; seeing how folks apply these and how they motivate their applications is really informative, something you cannot miss. We often won't directly discuss the readings, but don't let that imply they're not important. If I get the sense we're not keeping up with reading, expect the syllabus to change to incorporate quizzes or other accountability measures.

Reading for the course will consist of several books and articles (listed by week below). The books listed below also have Amazon links - you'll find most of these cheaper used online.

Required

- Box-Steffensmeier, Janet and Jones, Brad. 2004. Event History Modeling. Cambridge. ISBN 0521546737
- J. Scott Long. 1997. Regression Models for Categorical and Limited Dependent Variables. Sage Publications Inc. ISBN 0803973748
- Ward, Michael D. and John S. Ahlquist. 2018 Maximum Likelihood for Social Science. Cambridge. ISBN 978-1316636824.

Recommended

Useful, but not required (though some required reading in the first one):

- Gary King. 1998. Unifying Political Methodology. University of Michigan Press. ISBN 0472085549

Gary King's book is regarded as seminal in developing ML applications in political science. Scott Long's is a similarly accessible treatment of a host of ML models and applications (and the Stata book is a great applied companion). Together, these two books are probably the most important on the syllabus as they are both accessible, but comprehensive and technical enough to be useful. Ward and Ahlquist's book is a newer overview of applied ML in a political science setting. Box-Steffensmeier and Jones is a thorough and accessible treatment of hazard models in a variety of empirical settings.

Additional Resources

Other useful books include:

- Cameron, A. Colin and Trivedi, Pravin K. 1998. Regression Analysis of Count Data. Cambridge. ISBN 0521635675
- Maddala, Gregory. 1983. Limited Dependent and Qualitative Variables in Econometrics. Cambridge. ISBN 0521338255
- Paul D Allison - Event History Analysis : Regression for Longitudinal Event Data. Sage Publications Inc. ISBN 0803920555
- Tim Futing Liao - Interpreting Probability Models : Logit, Probit, and Other Generalized Linear Models. Sage Publications Inc. ISBN 0803949995
- John H Aldrich and Forrest D Nelson - Linear Probability, Logit, and Probit Models. Sage Publications Inc. ISBN 0803921330
- Fred C Pampel - Logistic Regression : A Primer. Sage Publications Inc. ISBN 0761920102
- Vani Kant Borooah - Logit and Probit : Ordered and Multinomial Models. Sage Publications Inc. ISBN 0761922423
- Scott R Eliason - Maximum Likelihood Estimation : Logic and Practice. Sage Publications Inc. ISBN 0803941072
- Richard Breen - Regression Models : Censored, Sample Selected, or Truncated Data. Sage Publications Inc. ISBN 0803957106
- Krishnan Namboodiri - Matrix Algebra : An Introduction. ISBN 0803920520

Course Requirements and Grades

The seminar requires the following:

- Problem sets - 60% total
- Mechanism papers - 40%

Please note that all written assignments must be submitted as PDFs either compiled in *LaTeX* or in R markdown (Quarto).

You'll complete a series of problem sets, mostly applied. How many will depend on how things move along during the term (roughly 8-10). Regarding the problem sets - the work you turn in for the problem sets should clearly be your own, but I urge you to work together - doing so is a great way to learn and to overcome problems.

The mechanism papers are a series of three short papers you'll write during the semester aimed at learning to identify and describe causal mechanisms, then at producing a causal mechanism. More on these early in the term.

A word about completeness - attempt everything. To receive a passing grade in the course, you must finish all elements of the course, so all problem sets, all exams, papers, etc. To complete an element, you must at least attempt all parts of the element - so if a problem set has 10 problems, you must attempt all 10 or the assignment is incomplete, you've not completed every element of the course, and you cannot pass. I realize there may be problems you have trouble with and even get wrong, but you must try - the bottom line is don't turn in incomplete work.

Grades will be assigned on the following scale:

Grade	Range	Grade	Range
A	94-100%	C+	77-79%
A-	90-93%	C	73-76%
B+	87-89%	C-	70-72%
B	83-86%	D	60-69%
B-	80-82%	F	<60%

Course Policies

Attendance

Attendance is expected, and is essential if you're to succeed in this class.

Academic Integrity

Ideas are the currency in academic exchange, so acknowledging where ideas come from is important. Acknowledging the sources of ideas also helps us identify an idea's lineage which can be important for understanding how that line of thought has developed, and toward promoting future growth. As graduate students, you should have a good understanding of academic honesty and best practices. Here are details of Binghamton's honesty policy.

Artificial Intelligence and Coding Tools

Generative AI can act as an incredible accelerator for research, for coding and data management in particular. Such tools are only as useful as your own mastery of coding, data management, and modeling.

Do you need to be able to write code when AI can do it for you? Yes. Three reasons:

1. *Understanding.* Code and the act of coding are the windows to seeing and understanding data, models, and predictions — how data produce models, which in turn produce predictions. Code is an explicit connection between a data matrix and a model, and facilitates your understanding of the what the model produces. AI can't hand you that understanding; you build it by doing the work yourself.
2. *Criticism.* AI will produce code. If you can code, you can read it, critique it, and evaluate its output to judge whether it's right. If you can't code, you can't do any of that — you're left trusting output you have no way to check.
3. *Inquiry.* If you don't understand the code, you probably don't understand the data and the model, and you'll not write a very good query to the AI in the first place. AI is only as good as its user. You should always know things the AI does not, in terms of your argument, your data, your model, and your code.

So here is the policy for this course. Do not use generative AI to produce code. If you get stuck coding, you may use AI to get *unstuck* - this is equivalent to consulting "stackexchange" or "stackoverflow" to solve code problems and get unstuck. Anytime you consult AI, you must cite and annotate it specifically: what you asked, why you asked it, and which parts of your code came from the AI. To this end, expect assessment where I ask you to show that you understand the data, code, models, and predictions you are producing. A bot can't fake that understanding.

You should not use generative AI tools for writing in this course. Learning to express your ideas about politics, about causal relations, about data and models, and explaining your findings are core skills for the Ph.D. Please learn to do these things without any use of generative AI.

Course Schedule

We meet Wednesdays. Three dates are dark: fall break (Oct 14), Nov 4 (I'm away at a conference), and Thanksgiving (Nov 25).

Part I — Foundations

Week 1, Aug 19 – Binary y Variables: probit/logit, first estimation, and interpretation

Required

- Ward & Ahlquist, 2018. *Maximum Likelihood for Social Science.* Chapters 1, 2, 4
- J. Scott Long. 1997. *Regression Models for Categorical and Limited Dependent Variables.* Chapter 3.

Recommended

- Hanmer and Kalkan (2013)
- Mood (2010)
- Pierskalla and Hollenbach (2013)

Week 2, Aug 26 – Likelihood Theory: the log-likelihood, score, Hessian, information

Required

- Gary King. 1998. *Unifying Political Methodology.* Chapters 1–4
- J. Scott Long. 1997. Chapters 1–2.

Recommended

- Eliason (1993)

Week 3, Sep 2 – MLE Mechanics and Building Your Own Model: grid search, Newton–Raphson, `optim`, and programming an estimator

Required

- Ward & Ahlquist, 2018. Chapters 5, 6, 7

Part II — Binomial extensions, and a causal lens

Week 4, Sep 9 – Binomial Extensions I — Diagnostics, Classification, and Fit: quantities of interest (review), symmetry, classification and ROC, separation, rare events

Required

- Ward & Ahlquist, 2018. Chapter 6 (quantities of interest)
- Gary King. 1998. Chapter 4 (simulation of quantities of interest)
- J. Scott Long. 1997. Chapter 5.
- G. King and Zeng (2001)
- Rainey (2016b)

Recommended

- Nagler (1994)
- Rainey (2024)
- C. Zorn (2005)
- Clark, Nordstrom, and Reed (2008)
- Gary King and Roberts (2015)

- Greenhill, Ward, and Sacks (2011)
- Esarey and Pierce (2012)
- Ward, Greenhill, and Bakke (2010)
- Rainey (2014)

Week 5, Sep 16 – Binomial Extensions II — Discrete-Time Hazards: the logit as a hazard, person-period (BTSCS) data, time dependence

Required

- Ward & Ahlquist, 2018. Chapter 11
- Beck, Katz, and Tucker (1998)
- Carter and Signorino (2010)

Recommended

- Metzger and Jones (2022)
- Wilson and Butler (2007)
- Svolik (2008)

Week 6, Sep 23 – Binomial Extensions III — Interactions: multiplicative terms, compression, and marginal effects in nonlinear models

Required

- Brambor, Clark, and Golder (2006)
- Berry, Golder, and Milton (2012)
- Berry, DeMeritt, and Esarey (2010)
- Hainmueller, Mummolo, and Xu (2019)
- Zhirnov, Moral, and Sedashov (2023)

Recommended

- Mize (2019)
- Rainey (2016a)
- Liu, Liu, and Xu (2025)

Week 7, Sep 30 – Causal Mechanics and Identification: potential outcomes, estimands (ATE/ATT), propensity scores, g-computation

Required

- Keele (2015)
- (Hernán and Robins 2020, chaps. 1–3 and 13) (potential outcomes, the g-formula)

Recommended

- Gary King and Nielsen (2019)

- Iacus, King, and Porro (2012)
- Blackwell (2013)
- Roberts, Stewart, and Nielsen (2020)
- Kam and Palmer (2008)
- Henderson and Chatfield (2011)

Part III — Counts and durations as one family

Week 8, Oct 7 – Event Counts I: Poisson, rates and offsets, equidispersion

Required

- Ward & Ahlquist, 2018. Chapter 10
- J. Scott Long. 1997. Chapter 8.1–8.2.
- Fordham (1998)

Recommended

- Gowa (1998)
- Brandt and Williams (2001)
- Fogarty and Monogan (2014)

Oct 14 – *no class, fall break*

Week 9, Oct 21 – Event Counts II: dispersion and zeros — negative binomial, zero-inflation, split populations

Required

- J. Scott Long. 1997. Chapter 8.3–8.7.
- Clark (2003)
- Bagozzi (2015)
- Casiraghi and Cusumano (2024)

Recommended

- C. J. W. Zorn (1998)
- Bagozzi et al. (2015)

Weeks 10–11, Nov 4 and Nov 11 – Continuous-Time Hazards and the Count $\leftrightarrow$ Hazard Equivalence: exponential, Weibull, and Cox models, and each as a Poisson

No meeting Nov 4 – read the notes on your own; we discuss them Nov 11.

Required

- Janet Box-Steffensmeier and Brad Jones. 2004. *Event History Modeling.* Chapters 1–5
- Box-Steffensmeier, Arnold, and Zorn (1997)
- Alt, King, and Signorino (2001)
- C. J. W. Zorn (2000)

Recommended

- Licht (2011)
- Kropko and Harden (2020)
- Harden and Kropko (2019)
- Box-Steffensmeier, De Boef, and Joyce (2007)
- Bagozzi et al. (2019)

Part IV — Choice, and synthesis

Week 12, Nov 18 – Choice I: multinomial and conditional logit, and prediction

Required

- J. Scott Long. 1997. Chapter 6.
- Ward & Ahlquist, 2018. Chapter 9
- Alvarez and Nagler (1998)
- Paolino (2021)

Recommended

- Hainmueller, Hopkins, and Yamamoto (2014)
- Signorino (2003)

Week 13, Dec 2 – Choice II: IIA — assessing it, and how much to worry; alternatives (multivariate probit) and the conditional–fixed-effects equivalence

Required

- Dow and Endersby (2004)
- Clark and Reed (2005)

Recommended

- Lacy and Burden (1999)
- C. J. W. Zorn (1996)
- Glasgow (2001)
- Beck (2020)
- Abramson, Koçak, and Magazinnik (2022)

Nov 25 – *no class, Thanksgiving*

References

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Bagozzi, Benjamin E., Minnie M. Joo, Bomin Kim, and Bumba Mukherjee. 2019. "A Bayesian Split Population Survival Model for Duration Data with Misclassified Failure Events." *Political Analysis* 27 (4): 415–34.

Beck, Nathaniel. 2020. "Estimating Grouped Data Models with a Binary-Dependent Variable and Fixed Effects via a Logit Versus a Linear Probability Model: The Impact of Dropped Units." *Political Analysis* 28 (1): 139–45.

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Berry, William D., Matt Golder, and Daniel Milton. 2012. "Improving Tests of Theories Positing Interaction." *Journal of Politics* 74 (3): 653–71.

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